

Day 3 — *Nimba* (Neem)

Module: Auśadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *nimba* (*Azadirachta indica*) by leaf shape, describe its dominant taste as *tikta* (bitter), name azadirachtin as its principal active compound, and explain why concentrated neem is **not** safe for ingestion at any age.

Materials

- A few dried neem leaves in a sealed container (smelling only — no chewing)
- A picture or — if you can — a neem twig with the characteristic compound leaf
- A small bottle of neem oil **kept by the teacher** (do not pass to students)
- A piece of neem-containing soap (commercial — most Indian general stores stock these) for the “what’s it actually in” demonstration
- Whiteboard

Vocabulary introduced today

- *nimba* (IAST: nimba; lit. “the bitter one”) — *Azadirachta indica*, the neem tree.
- *tikta* (IAST: tikta; lit. “bitter”) — one of the six tastes in Ayurveda; neem is the canonical *tikta* plant.
- *azadirachtin* — a complex limonoid compound. The most-studied bioactive in neem; the molecule behind neem’s use as an agricultural insect repellent.

Lesson flow

Warm-up (5 min)

Daily check-in: who has seen a neem tree near their home, school, or daily route? Hands.

“Anyone here ever been told ‘eat a neem leaf, it’s good for you’? Did you?” (Most who tried will say it tasted terrible.)

Bridge: today we explore why the body’s *first response* to neem — “spit it out” — is actually telling us something useful.

Core (20 min)

1. **Identify it.** Neem is a tree, not a small herb:
 - **Compound leaf:** instead of a single blade, each “leaf” is made up of 9–15 small leaflets attached to a central stem. (Compare to tulsi yesterday — single simple leaves.)
 - **Leaflet shape:** narrow, curved (sickle-shaped), with serrate (toothed) edges.
 - **Smell when crushed:** strong, slightly garlic-like, mostly bitter.

- **Where it grows:** native to South Asia; widely planted along streets, in temple compounds, and as a shade tree.
2. **The taste.** Pass the sealed container around for smell only.

Today we do not chew the leaf.

Neem is the textbook example of *tikta* (bitter taste) in Ayurveda. The bitter taste comes from a class of compounds called **limonoids** — the most-studied of which is **azadirachtin**. Bitter taste is the body's warning signal — many plant toxins are bitter, and the bitter receptors evolved to make us *cautious* about new plants.

3. **What azadirachtin does.** This is real:
- **Insecticidal/insect repellent:** azadirachtin disrupts insect moulting hormones. Hence neem oil is widely sold as an agricultural insect repellent — and that use is well-supported.
 - **Antimicrobial effects** at lab/petri-dish level.
 - **Traditional uses:** twigs as natural toothbrushes (*dātun*), neem-leaf paste on skin in some folk traditions, neem-based soaps and shampoos. These external/dilute uses are different from internal use.
4. **The safety warning — direct.** Write on the board:

⚠ **Neem is hepatotoxic at high doses.** Concentrated neem (especially neem oil ingested) has documented cases of acute liver injury, particularly in children. Neem oil is **not** for eating, drinking, or putting in anyone's mouth.

This is a real, documented safety concern in case-report literature. Neem leaf in small amounts (a few leaves brewed as a bitter tea in some traditions) is generally tolerated by adults — but it is **not** what we are teaching here, and self-prescription is not appropriate.

5. **What's it in, that we DO use?**

- Neem-based soap, shampoo, toothpaste — fine for external use as marketed.
- Neem oil for plants — used as agricultural pest control. Read the label.
- Neem twigs as toothbrushes — a long-standing dental tradition with modern dentistry's mixed view (some plaque-reducing benefit; abrasive enamel concern if used too often).

Activity (15 min) — Formative Quiz Part A + Compound-leaf sketch

First 5 min — Formative Quiz Part A (from quizzes/formative.md). Hand out the 5-question slip. Pencils only. Collect at minute 5.

Next 10 min — Compound-leaf sketch. - Each student draws a neem compound leaf from observation (or from the photo if no live sample). Label the rachis (central stem), the leaflets, the serrate edges. - Below the drawing, label: - **Sanskrit name:** *nimba* - **Botanical name:** *Azadirachta indica* - **Taste class:** *tikta* (bitter) - **Active compound:** azadirachtin - **Safety:** *not for ingestion in concentrated form*

Wrap-up (5 min)

- Restate safety rule + the neem-specific warning chorally.
- Hand-back of Day 2 exit tickets if marked.
- Preview Day 4: “*Tomorrow — the herb that turns everything gold.*”

Student-facing content

Nimba (*Azadirachta indica*) — known in English as neem.

How to identify it: - A tree, not a small plant. - **Compound leaf** — what looks like one “leaf” is actually made of 9–15 small leaflets arranged along a central stem. - Leaflets are narrow, slightly curved, with jagged edges. - Smell when crushed: bitter, slightly garlic-like.

What’s inside: Azadirachtin — a limonoid compound. The molecule behind neem’s well-supported use as an agricultural insect repellent.

Taste class in Ayurveda: *tikta* — bitter. Bitter taste is the body’s warning signal for many plant toxins.

Safety: - External uses (soap, oil for plants, twig as toothbrush) — fine when used as labelled. - **Concentrated internal use — especially neem oil — is unsafe.** Documented cases of liver toxicity in children. - We do not chew, drink, or eat neem in this class.

Homework

- Find a neem tree near your home or school. Sketch a leaf in your workbook from life. Note where the tree is and how many you saw on your walk.

Differentiation

- **One tier up:** Read up on the *Margosa oil poisoning* literature (margosa is another name for neem oil). Find one case-report finding. Be ready to share next class.
- **One tier down:** Focus on identifying the compound-leaf shape. The chemistry (azadirachtin) can come on Day 6 once you’ve seen more herbs.

Teacher notes

- The “don’t chew the leaf” rule is critical. A few brave/curious students will try if leaves are accessible — keep the container sealed and store it where students can’t reach during break.
- The hepatotoxicity warning needs to land as **specific knowledge**, not vague caution. Use the word “liver” explicitly. Some students have grandparents who give them neem leaves “for blood purification” — the lesson is to be respectful and curious about the traditional use *while* being clear about the dose concern.
- If a student asks “but if azadirachtin is dangerous, why have people used neem for centuries?” — the answer is: dilute external use and very small dietary use have a long history of being tolerated; *concentrated* use (oil, multi-leaf daily intake in adults, anything in children) is where the harm shows up.

Citation(s) used in this lesson

- *Azadirachta indica* and azadirachtin as principal limonoid: standard ethnobotanical and pharmacognosy textbook content.
- Neem oil hepatotoxicity: documented in published case reports in pediatric toxicology literature. We do not cite a specific paper here — teachers who want to deepen this should search “margosa oil hepatotoxicity case report” in PubMed.